BASINSHIFT

THE GLOBAL FLOOD INTELLIGENCE CHALLENGE

Full Challenge Documentation

Research rationale • Official problem statement • Dataset strategy • ML benchmark • Evaluation • Competition operations

Working title: BASINSHIFT
Challenge theme: Generalizable machine learning for extreme river-discharge prediction in unseen catchments
Document type: Organizer master specification with participant-facing statement
Version: 1.0 — Draft for technical review
Date: 3 November 2026

CONFIDENTIALITY NOTE

This document contains both proposed public challenge content and organizer-only benchmark design recommendations. Hidden-test construction, exact test catchments, evaluation labels, thresholds and anti-cheating controls must not be disclosed to participants before the competition closes.

Prepared as a design specification. It does not claim that the dataset package, hidden evaluator, baseline models or operational flood-warning system have already been built or independently validated.

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How to read this document: Sections 1–5 explain the competition concept and public challenge. Sections 6–14 specify the technical benchmark and competition rules. Sections 15–18 are intended primarily for organizers and reviewers.

1. Executive summary

BASINSHIFT is a proposed international machine-learning challenge focused on forecasting river discharge and detecting high-flow events in geographically unfamiliar catchments. The central research question is whether an ML system can learn transferable hydrological relationships rather than relying on patterns specific to catchments observed during training.

Participants will receive approved historical hydrological observations, meteorological inputs and catchment attributes. They will build systems that produce future discharge forecasts, probabilities of crossing high-flow thresholds and uncertainty estimates. Evaluation will include ordinary forecasting, rare-event detection, geographic generalization, calibration, robustness and computational efficiency.

The challenge is deliberately framed as a benchmark for generalization under environmental and geographic shift. It is not a request to build a complete flood-warning service, estimate street-level inundation, or replace official hydrological agencies.

Recommended direction

Proceed to benchmark engineering, not immediate public launch. The concept is promising, but launch readiness depends on dataset licensing and quality checks, aligned target/input definitions, leakage-resistant splits, baseline reproduction, hidden-evaluator testing and an independent pilot.

Key design principles

- Use genuine, documented hydrological observations for evaluation.
- Separate catchment identities and time periods to test distinct forms of generalization.
- Treat discharge forecasting, high-flow exceedance and inundation mapping as different tasks.
- Reward calibrated uncertainty and robust performance, not only average prediction error.
- Make the benchmark architecture-neutral and computationally accessible.
- Keep hidden labels, exact test partitions and test-generation details confidential.

Status of claims

Status label	Meaning in this document
Established resource	A named dataset or research resource that exists; exact coverage and permitted use still require verification.
Design proposal	A recommended competition choice that the organizing committee must implement and validate.
Unverified assumption	A point that must not be represented as fact until checked against the actual downloaded data or official documentation.
Out of scope	A capability the challenge does not claim to establish.

2. Challenge identity and scope

Working title: BASINSHIFT — The Global Flood Intelligence Challenge.

The name refers to the core technical challenge: learning across basin boundaries and operating under distribution shift. The title is provisional and should undergo trademark, domain and event-brand checks before publication.

2.1 Challenge category

Primary category: Machine Learning / Environmental Intelligence / Hydrological Forecasting. Secondary themes include time-series modeling, domain generalization, probabilistic prediction, transfer learning and robust ML.

2.2 Core objective

Develop a reproducible ML system that predicts river discharge at specified future horizons and estimates the probability of high-flow and extreme high-flow threshold exceedances, including for catchments excluded from model training.

2.3 In scope

- Time-series prediction of river discharge.
- High-flow threshold exceedance probability estimation.
- Generalization to catchments not seen during training.
- Uncertainty estimation and calibration.
- Robustness to missing, noisy or degraded input observations.
- Efficient and reproducible model inference.

2.4 Out of scope

- Certified public flood-warning decisions.
- Street-level flood depth or building-level inundation prediction in the core track.
- Claims of causal climate attribution.
- Autonomous emergency response or evacuation recommendations.
- Replacement of official hydrological forecasts or local expert judgment.

2.5 Intended participants

The challenge is intended for university teams, independent researchers, ML engineers, data scientists and multidisciplinary teams. The benchmark should be usable by teams with CPU-only or modest GPU resources; compute-intensive extensions may be optional rather than mandatory.

3. Problem background and motivation

River discharge is the volume of water flowing through a river cross-section per unit time. Forecasting discharge is a key component of hydrological monitoring and can support decisions in flood preparedness, reservoir operations, agriculture, infrastructure management and water-resource planning.

Catchments differ in climate, topography, soil, vegetation, land use, geology, river network structure and human intervention. A model trained on one set of basins may not transfer reliably to another. Temporal changes, rare extreme events and gaps in observation records further complicate learning.

A benchmark that evaluates only random samples from familiar catchments can overestimate real-world generalization. Overlapping time windows, shared catchment characteristics and temporally adjacent records may allow a model to perform well without learning transferable relationships.

3.1 Why this matters internationally

Flood hazards cross national and administrative boundaries, while hydrological data availability and monitoring quality vary widely. A generalizable approach could be useful in data-sparse basins, but that possibility must be demonstrated rather than assumed. A global benchmark can expose where models transfer, where they fail and how uncertainty changes under unfamiliar conditions.

3.2 The precise problem

Given a permitted history of hydrological and meteorological observations plus static or slowly varying catchment attributes, estimate river discharge at one or more future forecast horizons. For each horizon, also estimate the probability that discharge exceeds a pre-defined catchment-specific high-flow threshold.

3.3 Why the challenge is difficult

- Hydrological response is nonlinear and depends on interactions among rainfall, soil, terrain and catchment storage.
- Catchments have different response times and physical characteristics.
- Extreme events are relatively rare and may be underrepresented in training data.
- Observation records can be incomplete, noisy or inconsistent.
- Geographic and temporal shifts can invalidate learned correlations.
- Point forecasts alone do not communicate the risk of extreme outcomes.

3.4 Intended impact pathway

A successful research result may inform future decision-support systems, especially where model transfer and uncertainty are important. Before operational use, the system would require local validation, expert oversight, reliable data feeds, monitoring, and formal evaluation against relevant agency standards.

4. Research gap and ML justification

The competition is not based on the claim that hydrological forecasting is unsolved. Existing hydrological models, operational forecasting systems and ML methods already provide important capabilities. The proposed opportunity is to evaluate and improve reliability when models are transferred to unseen catchments and unusual conditions.

4.1 Research questions

1. How well can a model forecast discharge in catchments excluded from training?
2. Which static catchment attributes and dynamic forcing variables support transfer?
3. How does performance change between familiar and unfamiliar geographic regions?
4. Can probabilistic forecasts remain calibrated during high-flow events?
5. How sensitive are models to missing, noisy or delayed observations?
6. Do hybrid or physics-informed approaches improve generalization enough to justify their complexity?

4.2 Why machine learning is justified

The task involves learning nonlinear relationships across multiple input variables and catchments, handling heterogeneous time series, transferring representations across domains and estimating conditional predictive distributions. ML offers a family of methods suited to these tasks. However, the competition must permit statistical and physical baselines so that ML is required to demonstrate value rather than assumed to be superior.

4.3 Research hypotheses to test

Hypothesis	Testable evidence
H1: Catchment attributes improve transfer.	Compare otherwise identical models with and without static catchment attributes on unseen basins.
H2: Geographic holdouts reveal generalization gaps.	Compare familiar-basin temporal testing with fully unseen-basin testing.
H3: Probabilistic models improve risk communication.	Measure Brier score/CRPS, reliability and interval coverage.
H4: Robust training improves degraded-input performance.	Apply fixed missingness/noise perturbations and compare relative degradation.
H5: Hybrid approaches may improve physical plausibility.	Evaluate against the same hidden data and documented physical constraints.

These are hypotheses, not established findings. The competition should publish the hypotheses as motivation without implying that any particular method is guaranteed to win.

5. Official participant-facing problem statement

BASINSHIFT: Learning to Predict Extreme River Events Beyond Observed Environments

Background

Flood-related decisions depend in part on understanding how rivers respond to rainfall and changing environmental conditions. Yet hydrological systems differ substantially from one catchment to another. Models that perform well in familiar locations may not maintain the same reliability in geographically unfamiliar basins or during unusual events.

The challenge

Build a machine-learning system that forecasts future river discharge and estimates the probability of high-flow events across diverse catchments. Your solution must demonstrate generalization beyond the catchments used for training and provide evidence about the reliability of its predictions.

Core requirements

- Produce river-discharge forecasts for the official forecast horizons.
- Estimate high-flow and extreme high-flow exceedance probabilities.
- Demonstrate performance on unseen catchments.
- Quantify uncertainty using a documented method.
- Handle missing or degraded observations according to the benchmark specification.
- Provide reproducible code, model artifacts and technical documentation.

Potential input data

The organizer-provided package may include historical discharge, meteorological observations or forcing, catchment attributes, timestamps and data-quality indicators. The final data dictionary will specify the exact available variables, units, missing-value conventions and permitted use.

Expected output

For each evaluated catchment and forecast origin, teams will submit predicted discharge for each official horizon, high-flow exceedance probabilities, uncertainty information and required metadata in the published schema.

Permitted methods

Teams may use statistically justified approaches, tree-based models, neural sequence models, graph models, probabilistic methods, transfer learning, domain adaptation, physics-informed methods or hybrid systems. No specific architecture is mandatory. All external data, pretrained models and synthetic-data use must be declared.

Success

An exceptional solution will combine strong forecast skill with reliable high-flow detection, transfer to unseen catchments, calibrated uncertainty, robustness and reproducible engineering. The competition does not certify a system for operational flood warnings.

Official wording must be updated after the organizer has frozen the forecast horizons, variable list, data policy, split definitions and submission schema.

6. Dataset and data governance strategy

6.1 Primary resource: Caravan

Caravan is a large-sample hydrology dataset designed to make data from multiple catchment datasets more accessible in a standardized structure. The challenge proposal uses it as the primary starting point because it combines streamflow observations, meteorological forcing and catchment attributes across many catchments.

The often-cited collection size is 6,830 catchments. This figure must be checked against the exact release, version and files selected by the organizers. It must not be treated as a guarantee that every variable or every date is complete for every catchment.

6.2 Secondary and complementary sources

Resource	Potential role	Important caveat
GloFAS	Context, comparison or secondary forecast reference.	Check product version, licensing, spatial resolution, temporal coverage and whether its variables overlap with targets.
Copernicus Emergency Management Service	Optional spatial event-mapping extension.	Flood extent products are not interchangeable with river discharge observations.
Additional meteorological products	Fill or extend permitted forcing inputs where justified.	Ensure temporal/spatial alignment, licensing and no hidden-label leakage.

6.3 Data inventory required before release

- Dataset name, version, source URL and retrieval date.
- License and permitted competition, redistribution and commercial-use terms.
- Catchment identifier and geographic metadata.
- Variable names, units, measurement definitions and time zones.
- Observation frequency, date coverage and missingness rate.
- Quality flags, measurement uncertainty and known caveats.
- Spatial resolution and mapping between sources.
- Data transformations, filters and derived features.
- Checksum or immutable manifest for released files.

6.4 Data processing rules

- Preserve raw downloaded data separately from processed benchmark data.
- Record every transformation in a version-controlled pipeline.
- Normalize units explicitly and test conversions.
- Use UTC or a clearly documented canonical time convention.
- Do not forward-fill across long gaps without an explicit rule.
- Never replace missing discharge labels with zero.
- Fit scalers, imputers and feature-selection steps using training data only.
- Record catchment-level exclusions and their reasons.

6.5 Synthetic data policy

Synthetic data may be used for training or augmentation if disclosed and documented. Synthetic data must not be passed off as measured observations and must not replace genuine hidden evaluation labels. Organizers should define whether external pretraining is allowed, whether public external labels may be used and how data provenance is audited.

6.6 Privacy and licensing

Hydrological data are often environmental rather than personal data, but licensing, critical-infrastructure sensitivity and location-related restrictions still require review. Organizers must verify rights to redistribute every dataset component and must not assume that public availability means unrestricted reuse.

7. Benchmark and split design

The benchmark's credibility depends on split design. The organizing committee should define splits at the catchment level before generating windows, sequences or samples. Randomly splitting overlapping windows is not sufficient for a geographic-generalization challenge.

7.1 Proposed evaluation partitions

Partition	Design	Access	Purpose
Training	Selected catchments and historical periods.	Public	Fit model parameters.
Validation	Separate catchments and/or time windows defined in advance.	Public	Model selection and iteration.
Test A: Temporal	Later periods in catchments represented in training, where feasible.	Hidden	Temporal extrapolation.
Test B: Geographic	Catchment identities fully excluded from training.	Hidden	Unseen-catchment transfer.
Test C: Stress	Controlled input corruption or distribution-shift cases.	Hidden	Robustness and reliability.

7.2 Leakage controls

- Assign catchment IDs to partitions before constructing training windows.
- Ensure no target labels from hidden periods enter training-derived statistics.
- Compute catchment thresholds from an approved historical reference period only.
- Fit preprocessing parameters using training data only.
- Audit external datasets for hidden-period or hidden-location labels.
- Inspect public mirrors and derived products for possible reconstruction of hidden targets.
- Check that forecast input timestamps never include information after the forecast origin.
- Run duplicate and near-duplicate record checks across partitions.

7.3 Geographic split strategy

Do not rely on a single random basin split. Consider region-stratified holdouts so that test partitions include a defensible mix of climatic and geographic conditions. Report the distribution of catchment area, aridity, elevation, climate zone and record length across splits. The exact strategy should be selected after coverage analysis.

7.4 Temporal split strategy

For temporal testing, use chronologically later periods than the training period. Avoid random time shuffling. Ensure that the input history available at each forecast origin is strictly prior to the target horizon. Document whether a warm-up context is permitted for a test catchment and whether that context contains labels.

7.5 Stress-test design

Apply reproducible perturbations such as fixed rates of missing observations, short contiguous gaps, measurement noise and delayed input availability. Publish the perturbation families and general methodology, but keep exact hidden-test seeds and examples confidential until after the competition.

7.6 Split acceptance criteria

The benchmark is not ready until independent reviewers confirm that split assignments are reproducible, the hidden targets cannot be trivially reconstructed, the partitions are sufficiently large for stable metrics and no accidental identity or time leakage remains.

8. ML task and input/output contract

8.1 Task formulation

For catchment c and forecast origin t, the model receives a permitted input history X(c, ≤t) and static or slowly varying attributes A(c). It predicts discharge q(c,t+h) at each official horizon h, plus the probability of exceeding the official threshold T(c).

The forecast horizons should be selected after inspecting the temporal resolution, label coverage and practical baseline difficulty. A provisional set of 1, 3 and 7 days may be explored only if the data support those horizons; it is not a frozen requirement.

8.2 Input fields

Field group	Example fields	Rules
Identity	Catchment ID, forecast origin timestamp.	IDs are identifiers, not hidden labels.
Hydrology history	Past observed discharge values and quality flags.	Only observations available at forecast origin.
Meteorology	Past and permitted forecast precipitation, temperature or other forcing.	Clearly distinguish observed history from forecast inputs.
Catchment attributes	Area, elevation, soil/land attributes where available.	Static fields must be documented and checked for leakage.
Missingness metadata	Masks, quality flags, observation gaps.	Use only if included in the official contract.

8.3 Output schema (illustrative)

The following is a conceptual example only. The organizers must publish a formal JSON Schema or equivalent machine-readable contract and define units, probability semantics and uncertainty representation.


```
{ "catchment_id": "BASIN_001", "forecast_origin": "2026-01-15T00:00:00Z", "forecast_horizons_days": [1, 3, 7], "predicted_discharge": [12.4, 14.1, 18.7], "high_flow_probability": [0.08, 0.17, 0.31], "extreme_flow_probability": [0.01, 0.04, 0.12], "prediction_intervals": { "lower": [8.1, 8.8, 9.5], "upper": [17.9, 22.6, 31.2] } }
```


All numeric values above are illustrative placeholders, not actual observations or recommended predictions.

8.4 Validation rules

- Reject unknown catchment IDs and unexpected forecast horizons.
- Reject NaN, infinity, invalid probability values and malformed records.
- Require non-negative discharge predictions unless a documented target transformation changes the output semantics.
- Check prediction interval ordering and required coverage fields.
- Enforce one prediction per required key and prevent duplicate submissions.
- Record missing outputs and apply a published penalty or submission rejection rule.

9. Modeling approaches and baselines

The competition should be architecture-neutral. Participants should be able to justify any method that satisfies the interface and constraints. Organizers should provide strong, reproducible baselines to establish task difficulty and reduce the risk that the benchmark is accidentally trivial or impossible.

9.1 Required organizer baselines

Baseline	Role	Implementation considerations
Persistence	Simple reference forecast.	Use last available discharge or a clearly defined persistence variant.
Seasonal/climatological	Reference for recurring patterns.	Fit using training data only; define handling of sparse records.
Gradient boosting	Strong tabular/time-lag baseline.	Document features, missingness handling, hyperparameters and compute.
Sequence model	Neural temporal baseline.	Use a modest recurrent or temporal convolution model; report seeds and runtime.
Optional hydrological/hybrid	Physical or process-based reference.	Include only if inputs, calibration and runtime are comparable.

9.2 Candidate participant methods

- Gradient-boosted decision trees using lagged forcing, discharge history and catchment attributes.
- Recurrent models such as LSTM or GRU for sequential dependencies.
- Temporal convolutional networks for multi-scale temporal patterns.
- Transformer-style sequence models where the data volume and compute justify them.
- Graph neural networks if river-network structure is reliably available and standardized.
- Probabilistic regression, quantile models or distributional forecasting.
- Domain adaptation, meta-learning or invariant representation methods.
- Physics-informed losses or hybrid physical/ML models, with constraints documented.

9.3 Model selection discipline

Teams should select methods based on the data regime, forecast horizon, compute budget and validation evidence. A complex model is not inherently superior. The final report should include ablations showing which components materially improve performance.

9.4 Compute fairness

Set an inference environment and resource budget before launch. Consider CPU inference as the baseline requirement, with a modest optional GPU allowance if the organizer platform supports it. The precise CPU, RAM, accelerator, runtime and artifact-size limits must be published after a pilot.

10. Evaluation methodology

The final score should combine complementary metrics. The metric implementation, aggregation rules, baseline normalization and tie-breaking rules must be frozen before submissions open.

10.1 Metric families

Dimension	Candidate metrics	What it measures
Continuous forecast	MAE, normalized MAE, NSE (with careful handling).	Magnitude of forecast errors and hydrological fit.
Event detection	AUPRC, event recall, precision at fixed alert budget.	Ability to identify rare high-flow events.
Operational alert burden	False alarms per catchment or per unit time.	Practical cost of excessive alerts.
Probabilistic quality	Brier score, CRPS, reliability diagrams.	Quality of event probabilities and predictive distributions.
Interval quality	Coverage and interval width.	Whether uncertainty intervals are reliable and informative.
Generalization	Macro-average by unseen catchment/region.	Transfer to unfamiliar domains.
Robustness	Relative degradation under controlled corruption.	Sensitivity to missing/noisy inputs.
Efficiency	Runtime, memory and artifact size.	Practical inference cost.

10.2 Proposed machine score

Component	Weight	Implementation note
Forecast skill	25	Combine predeclared continuous metrics after baseline-aware normalization.
High-flow event detection	25	Use event metrics appropriate for class imbalance and fixed alert budget.
Out-of-distribution generalization	20	Score unseen catchments and report region-level dispersion.
Calibration and uncertainty	15	Use proper scoring rules and interval reliability.
Robustness	10	Evaluate fixed, predeclared input corruption scenarios.
Efficiency	5	Measure under the standardized execution environment.
Total	100	Freeze exact formula and directionality before competition.

This is a proposed weighting scheme. It should be calibrated through baseline and pilot results. Metric aggregation must avoid double-counting the same underlying error signal.

10.3 Normalization

A defensible option is to map each metric to a bounded score using fixed baseline anchors and predeclared caps. For example, improvement over a baseline can be normalized against a fixed target range. The formula must handle negative skill, zero denominators, degenerate cases and missing outputs. Do not choose normalization after seeing participant results.

10.4 Statistical reporting

- Report both macro-average across catchments and pooled observation-level results.
- Use catchment-level bootstrap intervals where appropriate.
- Report event counts and denominators with every rare-event metric.
- Report performance by region, catchment characteristics and event severity.
- Record variance across random seeds for stochastic methods.
- Do not claim meaningful superiority when confidence intervals overlap materially.
- Publish failure cases and metric limitations after the competition.

10.5 Avoiding accuracy-only ranking

A model may achieve low average error while missing rare extreme events, or may issue too many false alarms. A probabilistic model may be better calibrated even if its point forecast is slightly less accurate. The scoring system must preserve these distinctions.

11. Competition rules and constraints

11.1 Data rules

- Use organizer-provided data according to the published license and terms.
- Declare all external datasets, pretrained weights, foundation models and synthetic data.
- Do not access, infer, scrape or reconstruct hidden evaluation labels.
- Do not use future observations or post-origin information as input.
- Do not manually edit predictions for hidden test records.
- Retain data provenance and transformation records.

11.2 Model and API rules

- Submissions must execute in the organizer-defined environment.
- External network access during final inference should be disabled unless explicitly permitted.
- Proprietary APIs must not be mandatory; if permitted, teams must disclose dependency and cost.
- Model weights and source code must be submitted in the required format.
- All inference must follow the official input/output contract.
- Teams must disclose model lineage and material third-party components.

11.3 Team and authorship rules

The organizing committee must define team size, eligibility, collaboration rules, open-source obligations, intellectual-property terms, attribution requirements and conflict-of-interest disclosures. These administrative terms are not determined by this technical document.

11.4 Submission attempts

Use a limited number of public leaderboard submissions to reduce overfitting. The number, reset policy, late-submission handling and final cutoff must be published before registration closes.

11.5 Disqualification conditions

- Attempted access to hidden labels or evaluation infrastructure.
- Undisclosed external data or prohibited inference-time network calls.
- Fabricated experiments, benchmark results or data provenance.
- Malicious code, environment escape attempts or resource abuse.
- Submission materially different from the model defended to judges.
- Violation of licensing, privacy or organizer-defined collaboration rules.

12. Submission deliverables

Deliverable	Minimum expected contents
1. Executable model	Inference code, weights/artifacts and entry point.
2. Source code	Readable repository, setup instructions and versioned commit.
3. Dependency manifest	Pinned packages, runtime requirements and environment definition.
4. Technical report	Problem framing, hypotheses, methods, experiments, results and limitations.
5. Data provenance	All sources, licenses, transformations and external data declarations.
6. Evaluation evidence	Public validation results, ablations and robustness analysis.
7. Reproduction guide	Commands to install, run inference and reproduce reported metrics.
8. Model card	Intended use, limitations, risks, training data and evaluation scope.
9. Demonstration	Short technical demo or presentation showing system behavior.
10. Team declaration	Authors, roles, conflicts and third-party contributions.

12.1 Suggested technical report structure

1. Abstract and executive summary.
2. Problem formulation and research hypothesis.
3. Dataset and feature provenance.
4. Preprocessing and leakage controls.
5. Model architecture and training procedure.
6. Baseline comparison and ablation study.
7. Temporal and unseen-catchment results.
8. Extreme-event and calibration results.
9. Robustness and efficiency analysis.
10. Limitations, failure cases and responsible-use considerations.
11. Reproduction instructions and artifact references.

12.2 Technical defense questions

- What evidence shows the model generalizes to catchments not used during training?
- Which input features would be unavailable in a real deployment?
- How are high-flow thresholds defined and frozen?
- How does the model behave when discharge history is missing?
- Which ablation most changed the result?
- How are uncertainty estimates calibrated?
- What are the largest failure modes?
- What would be required before deployment in a real basin?

13. Judging rubric

Category	Points	Evidence judges should inspect
ML performance and generalization	20	Hidden benchmark score, unseen-basin performance and event detection.
Research novelty	15	Specific research contribution, hypothesis and evidence beyond standard integration.
Technical architecture	10	Model rationale, data pipeline, system design and implementation.
Robustness and reliability	15	Stress tests, failure analysis, calibration and stability.
Real-world applicability	10	Deployment pathway, constraints and limitations.
Reproducibility and engineering	15	Code, pinned dependencies, provenance, repeatable runs and logs.
Presentation and technical defense	5	Clarity, evidence quality and answers to technical questions.
Uncertainty and responsible AI	10	Probabilistic quality, risk communication and responsible-use analysis.
TOTAL	100	

13.1 Score anchors

Score band	Interpretation
0–20% of category points	Absent, invalid or unsupported evidence.
21–40%	Weak execution; substantial methodological or engineering gaps.
41–60%	Competent baseline-level work with limited differentiation.
61–80%	Strong work with credible evidence and sound decisions.
81–100%	Exceptional, well-defended work with reproducible evidence and clear differentiation.

13.2 Judge calibration

Judges should independently score submissions before discussion. The committee should provide example submissions or calibration exercises, define conflict-of-interest procedures and record score justifications. Large scoring disagreements should trigger a review rather than informal averaging without explanation.

13.3 Novelty standard

Novelty should be judged against relevant prior work and the team's own evidence. A new combination of known methods may be useful engineering but should not automatically receive high research-novelty points. A smaller, well-supported contribution can be more valuable than an elaborate but unvalidated architecture.

14. Hidden evaluation and competition integrity

14.1 Threat model

Threat	Control
Catchment/time leakage	Split by catchment and chronology before window generation; audit overlap.
Hidden label reconstruction	Restrict test periods, inspect public data mirrors and maintain protected labels.
Leaderboard overfitting	Limit submissions; use public validation and final hidden evaluation.
Hardcoded outputs	Use unseen catchments/origins, randomized record ordering where safe and independent inference.
External data leakage	Require source declarations and audit feature provenance.
Fabricated results	Execute submitted code and reproduce key claims.
Environment abuse	Run isolated containers with resource and network controls.
Model substitution	Hash artifacts and verify the model used in the technical defense.

14.2 Evaluation execution

1. Freeze the dataset release, split manifest and metric package.
2. Build an isolated, versioned evaluator image.
3. Validate the submission schema before scoring.
4. Execute inference with fixed resource limits and controlled inputs.
5. Store logs, artifact hashes, environment versions and score outputs.
6. Run independent reproduction for finalists.
7. Escalate anomalies to a technical integrity panel.
8. Publish aggregate results and methodology after the competition.

14.3 Confidentiality boundaries

Participant-facing materials may describe the existence and purpose of hidden geographic, temporal and stress tests. They should not disclose exact hidden catchments, time windows, threshold values, stress-test seeds, private labels or implementation details that would enable direct reconstruction. Judges and evaluators should operate under access controls and conflict-of-interest rules.

15. Organizer implementation plan

The committee should treat benchmark creation as a software and research project with explicit acceptance gates. A polished problem statement is not a substitute for a functioning benchmark.

Phase	Workstream	Exit criteria
1. Research audit	Dataset/version/license verification; literature review; task definition.	Signed source register and feasibility report.
2. Data engineering	Download, validate, align, clean and version data.	Reproducible data pipeline and data dictionary.
3. Split construction	Create catchment and temporal partitions; audit leakage.	Approved split manifest and red-team report.
4. Baselines	Implement persistence, seasonal, tree and sequence baselines.	Reproducible metrics and resource profiles.
5. Evaluator	Build schema validation, isolated inference and scoring.	End-to-end test on sample submissions.
6. Pilot	Invite independent test teams; observe failure modes.	Pilot report and resolved critical issues.
7. Release	Publish statement, data, rules, starter kit and support docs.	Public launch checklist signed off.
8. Post-event	Reproduce finalists, publish results and benchmark paper/report.	Auditable final results and lessons learned.

15.1 Suggested technical components

Component	Possible implementation
Data pipeline	Python, pandas/polars, xarray where appropriate, Parquet/Zarr.
ML baselines	scikit-learn, XGBoost/LightGBM, PyTorch.
Experiment tracking	MLflow or an equivalent reproducible tracking system.
Submission interface	Versioned JSON Schema or typed API contract.
Execution	Docker or equivalent isolated container environment.
Scoring	Versioned metric package with unit and integration tests.
Artifact registry	Immutable storage with hashes and access controls.
Documentation	Version-controlled data dictionary, rules and benchmark guide.

These are suggested tools, not mandatory dependencies. Final selection should reflect data volume, platform constraints, participant access and organizer expertise.

15.2 Minimum staffing

- Hydrology/domain lead responsible for scientific validity and target definitions.
- ML benchmark lead responsible for splits, metrics and baseline quality.
- Data engineer responsible for ingestion, provenance and data quality.
- Platform engineer responsible for submission execution and isolation.
- Competition operations lead responsible for rules, participant support and schedule.
- Independent red-team reviewers responsible for leakage and fairness audits.

15.3 Pilot requirements

The pilot should include teams that did not build the benchmark. Ask them to reproduce baselines, train at least one alternative model, submit through the official interface and attempt to identify leakage or ambiguities. A pilot that only confirms the organizer's own code works is insufficient.

16. Risks, limitations and responsible use

Risk	Severity	Mitigation / disclosure
Global coverage is uneven	High	Publish catchment distribution and regions not represented.
Discharge is mistaken for inundation	Critical	Keep the core target explicit; do not imply street-level flood maps.
Extreme-event labels are sparse	High	Report event counts and uncertainty; avoid unstable rankings.
Measurement errors or missing records	High	Retain quality flags and disclose exclusions.
Geographic leakage	Critical	Independent split audit before release.
External data contaminates test	High	Source register and hidden-label reconstruction review.
Compute inequality	Medium	Standardized inference budgets and CPU-capable baselines.
Poor calibration	High	Score proper probabilistic metrics and report reliability.
Benchmark overfitting	High	Limited submissions and final hidden evaluation.
Operational misuse	Critical	Explicitly state that benchmark success is not deployment certification.

16.1 Responsible-use statement

BASINSHIFT is an experimental research benchmark. Model outputs must not be used as the sole basis for evacuation, emergency response, infrastructure operation or public warning decisions. Any future deployment requires local validation, hydrological expert review, appropriate data governance, operational monitoring, fail-safe procedures and coordination with competent authorities.

16.2 Known limitations of the proposed design

- Caravan's global coverage does not mean uniform coverage, complete records or identical data quality.
- Catchment discharge is a point or basin-level hydrological variable and does not directly describe local flood depth or exposure.
- High-flow percentile thresholds are statistical benchmark labels, not official hazard thresholds.
- Publicly available secondary datasets may have different spatial and temporal semantics.
- Benchmark performance does not establish causal understanding or operational readiness.
- Results may depend on the selected regions, periods, horizons and data versions.

17. Acceptance criteria and launch checklist

A public launch should be blocked until all critical checks pass. The committee should maintain evidence for each item rather than marking tasks complete based on verbal assurance.

Gate	Acceptance criterion	Status before verification
Scientific validity	Hydrology expert approves target, horizons and threshold semantics.	OPEN
Dataset rights	Every distributed file has verified license and redistribution permission.	OPEN
Data quality	Coverage, missingness, units and exclusions documented.	OPEN
Split integrity	Independent leakage audit completed.	OPEN
Baseline reproducibility	All required baselines rerun from clean environment.	OPEN
Metric correctness	Metric package unit-tested on synthetic edge cases and known examples.	OPEN
Submission interface	Valid/invalid example submissions tested end-to-end.	OPEN
Hidden evaluator	Isolation, resource limits, logs and access controls verified.	OPEN
Fairness	CPU/modest-compute path demonstrated and constraints published.	OPEN
Pilot	Independent teams complete full workflow successfully.	OPEN
Rules	Eligibility, IP, data, collaboration and appeals rules approved.	OPEN
Communications	Public statement accurately reflects what is implemented.	OPEN

17.1 Go/no-go criteria

GO: all critical gates pass, baseline difficulty is meaningful, the evaluator is reproducible, and independent reviewers approve the benchmark.

CONDITIONAL GO: only non-critical documentation or polish remains, with an owner and deadline; no hidden-test or licensing risk remains open.

NO-GO: unresolved licensing, label leakage, invalid target definitions, unrepeatable scores, hidden-test exposure or inability to reproduce final submissions.

17.2 Final committee recommendation

Proceed with BASINSHIFT as the leading challenge concept and begin a time-boxed feasibility and benchmark-engineering phase. Do not advertise the challenge as a validated global flood-prediction benchmark until the data pipeline, baseline suite, hidden evaluator and independent pilot have passed their acceptance gates.

18. References and source notes

The following resources are recommended starting points for verification and benchmark construction. The organizing committee must check the current release, license, data coverage and documentation directly before using or redistributing any resource.

Resource	Use in BASINSHIFT	Official / primary link
Caravan — global large-sample hydrology dataset	Primary candidate dataset for streamflow, meteorological forcing and catchment attributes.	https://doi.org/10.1038/s41597-023-01975-w
Global Flood Awareness System (GloFAS)	Potential operational context and secondary comparison resource.	https://global-flood.emergency.copernicus.eu/
Copernicus Emergency Management Service	Potential flood event mapping and optional spatial extension.	https://emergency.copernicus.eu/
WeatherBench 2	Reference for ML weather forecasting benchmark practices; not the core hydrology dataset.	https://weatherbench2.readthedocs.io/
NASA Prediction of Worldwide Energy Resources (POWER)	Potential meteorological/solar data source; suitability and licensing must be checked for the specific task.	https://power.larc.nasa.gov/
HydroShare	Potential repository for hydrological data and reproducible research artifacts.	https://www.hydroshare.org/

Source and verification note

This document is a competition design proposal synthesized from the selected BASINSHIFT concept and the technical requirements established in the preceding discussion. It is not a systematic literature review, legal opinion, hydrological validation report or confirmation that all listed sources can be combined for this task. Before publication, add a dated bibliography of peer-reviewed papers, dataset release notes, official license texts and exact data-version identifiers.

Recommended additional research before launch

- Review the Caravan paper, supplement, release documentation and source dataset licenses.
- Review GloFAS product documentation for target semantics, forecast lead times and access terms.
- Review relevant hydrological model intercomparison and cross-catchment ML literature.
- Review probabilistic hydrological forecasting and uncertainty calibration literature.
- Review benchmark leakage and data contamination risks for every candidate external source.
- Record all research decisions in a version-controlled benchmark design log.

END OF DOCUMENT

BASINSHIFT — Full Challenge Documentation | Version 1.0 Draft